

ACC Two-Channel Torque Assembly

Physical Conditions — Smoothstep Gate Inputs

Proximity $|\Delta x|$ · Quietness $\text{EMA}(|v|)$ · Stability $\theta, \dot{\theta}$ · Contact Fz^L, Fz^R · Terrain $h^L_{\text{gnd}}, h^R_{\text{gnd}}$

Wheel Torque τ_w

τ_{balance}
Sagittal Balance Core
(pitch PD + vel. damping
+ position P, per-wheel damping)
always active

+

$g_{\text{anchor}} \cdot \tau_{\text{anchor}}$
Proximity-Gated Anchor
(integral + damping boost)
gated by $g_{\text{prox}}, g_{\text{env}}, g_{\theta}$

$\oplus$ convex blend, not a sum

τ_{flight} (override)
Contact-Loss Recovery
(reaction-wheel attitude PD)
blended in by g_{flight}

$$\tau_w = (1 - g_{\text{flt}}) \cdot (\tau_{\text{bal}} + g_{\text{anc}} \cdot \tau_{\text{anc}}) + g_{\text{flt}} \cdot \tau_{\text{flt}}$$

Leg-Joint Torque τ_q

$\tau_{\text{posture}}(h^L_{\text{cmd}}, h^R_{\text{cmd}})$
Posture & Height Regulation
(Jacobian PD + hip-yaw
divergence + homing; per-leg
ground split via g_{terrain})

+

$\tau_{\text{lat}} + \tau_{\text{yaw}}$
Lateral & Heading Regulation
(antisym. hip-roll / hip-yaw
pairs, saturated)
always active

$$\tau_q = \tau_{\text{post}}(h^L, h^R) + \tau_{\text{lat}} + \tau_{\text{yaw}}$$

Channels are independent on level ground; across a step g_{terrain} also retunes the wheel loop